Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Manuel & Maria Silva Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
2800 White RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X100-X006-XXXX X057-X019-X004-XXXX

B. OPERATORS

Silva, Manuel & Maria

Operator name: Silva, Manuel & Maria

Telephone no.: Landline: ; Cellular: (209) 564-7297

Mailing Address Number and Street	City	State	Zip Code
21487 White	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Azevedo, Manuel & Maria

Legal owner name: Azevedo, Manuel & Maria

Telephone no.: Landline: (209) 632-6393; Cellular: (209) 535-0646

Mailing Address Number and Street	City	State	Zip Code
2800 White RD	Turlock	CA	95380

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	110	100
Number under roof	920	155	0	0	0	0
Maximum number	920	155	0	0	110	100
Average number	920	155	0	0	110	100
Avg live weight (lbs)	1,300	1,400	0	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 59 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,366.13 *tons per reporting period*

Total nitrogen from manure: 332,039.46 *lbs per reporting period*

After ammonia losses (30% loss applied): 232,427.62 *lbs per reporting period*

Total phosphorus from manure: 53,873.08 *lbs per reporting period*

Total potassium from manure: 169,539.65 *lbs per reporting period*

Total salt from manure: 468,824.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,655,000 *gallons*

Total nitrogen generated: 48,594.81 *lbs*

Total phosphorus generated: 4,635.78 *lbs*

Total potassium generated: 75,299.52 *lbs*

Total salt generated: 455,158.83 *lbs*

	11,655,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	11,655,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well	Ground water
Domestic Well	Ground water
TID Irr. Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	Scraped material	1,100.00 *ton*	Dry-weight	56.3		15,900.00	4,300.00	28,000.00		0.00
10/17/2023	Scraped material	1,100.00 *ton*	Dry-weight	69.4		26,600.00	5,590.00	8,770.00		43.80

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	33,193.38	7,897.21	32,823.16	294,861.60
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	33,193.38	7,897.21	32,823.16	294,861.60

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1	12	12	2	both	X045-X100-X006-XXXX
Field #2	80	80	4	both	X057-X019-X005-XXXX
Field #3	25	25	2	both	X045-X100-X051-XXXX
Field #4	14	14	2	both	X045-X120-X012-XXXX
Totals for areas that were used for application	131	131	10		
Totals for areas that were not used for application					
Land application area totals	131	131	10		

B. CROPS AND HARVESTS

Field #1

Field name: Field #1

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 12 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	183.60 *ton*	Dry-weight		75.0	22,900.00	4,600.00	55,100.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	172.50	34.50	207.00	0.00
Total actual harvest content	15.30	175.19	35.19	421.52	474.30

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	330.00 *ton*	Dry-weight		68.8	14,600.00	2,450.00	16,900.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	29.00	263.90	43.50	301.60	0.00
Total actual harvest content	27.50	250.54	42.04	290.00	1,149.72

Field #2

Field name: Field #2

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 80 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	612.00 *ton*	Dry-weight		64.6	19,800.00	4,900.00	52,500.00		11.20
04/27/2023	612.00 *ton*	Dry-weight		71.2	23,800.00	4,100.00	50,700.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	210.00	52.50	279.00	0.00
Total actual harvest content	15.30	212.11	44.61	507.75	1,157.41

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,100.00 *ton*	Dry-weight		70.0	13,400.00	2,050.00	14,600.00		4.70
09/13/2023	1,100.00 *ton*	Dry-weight		72.2	15,400.00	2,710.00	19,000.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	29.00	249.40	43.50	290.00	0.00
Total actual harvest content	27.50	228.28	37.63	265.71	854.10

Field #3

Field name: Field #3

Field #3

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 25 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	382.50 *ton*	Dry-weight		75.0	22,900.00	4,600.00	55,100.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	172.50	34.50	207.00	0.00
Total actual harvest content	15.30	175.19	35.19	421.52	474.30

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	687.50 *ton*	Dry-weight		68.8	14,600.00	2,450.00	16,900.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	29.00	263.90	43.50	301.60	0.00
Total actual harvest content	27.50	250.54	42.04	290.00	1,149.72

Field #4

Field name: Field #4

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 14 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	214.20 *ton*	Dry-weight		65.4	17,700.00	4,600.00	51,400.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	183.00	48.00	267.00	0.00
Total actual harvest content	15.30	187.40	48.70	544.20	1,164.64

Field #4

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	385.00 *ton*	Dry-weight		70.8	13,000.00	2,610.00	18,500.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	29.00	232.00	43.50	290.00	0.00
Total actual harvest content	27.50	208.78	41.92	297.11	1,043.90

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1 - 11/06/2022: Winter Forage

Field name: Field #1

Crop: Winter Forage — Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	89.94	21.72	87.61	1,151.02	120.00 *ton*
Application event totals		89.94	21.72	87.61	1,151.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	60.74	7.75	88.53	520.10	135,000.00 *gal*
Application event totals		60.74	7.75	88.53	520.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/23/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	60.74	7.75	88.53	520.10	135,000.00 *gal*
Application event totals		60.74	7.75	88.53	520.10	

Field #1 - 05/14/2023: Corn, silage

Field name: Field #1

Crop: Corn, silage — Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	4.38	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		4.38	0.00	0.00	0.00	

Field #1 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	131.68	12.87	216.55	1,361.90	360,000.00 *gal*
TID Irr. Canal	Surface water	3.94	0.00	0.00	0.00	1,620,000.00 *gal*
Application event totals		135.63	12.87	216.55	1,361.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.94	0.00	0.00	0.00	1,620,000.00 *gal*
Application event totals		3.94	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	93.76	6.92	146.77	884.99	315,000.00 *gal*
TID Irr. Canal	Surface water	2.63	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		96.38	6.92	146.77	884.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.29	0.00	0.00	0.00	1,350,000.00 *gal*
Application event totals		3.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	93.76	6.92	146.77	884.99	315,000.00 *gal*
TID Irr. Canal	Surface water	2.63	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		96.38	6.92	146.77	884.99	

Field #1 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.29	0.00	0.00	0.00	1,350,000.00 *gal*
Application event totals		3.29	0.00	0.00	0.00	

Field #2 - 11/06/2022: Winter Forage

Field name: Field #2

Crop: Winter Forage Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	89.94	21.72	87.61	1,151.02	800.00 *ton*
Application event totals		89.94	21.72	87.61	1,151.02	

01/25/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.67	6.98	79.68	468.09	810,000.00 *gal*
Application event totals		54.67	6.98	79.68	468.09	

03/24/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.67	6.98	79.68	468.09	810,000.00 *gal*
Application event totals		54.67	6.98	79.68	468.09	

Field #2 - 05/14/2023: Corn, silage

Field name: Field #2

Crop: Corn, silage Plant date: 05/14/2023

Field #2 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.19	0.00	0.00	0.00	8,730,000.00 *gal*
Application event totals		3.19	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	59.26	5.79	97.45	612.86	1,080,000.00 *gal*
TID Irr. Canal	Surface water	3.12	0.00	0.00	0.00	8,550,000.00 *gal*
Application event totals		62.38	5.79	97.45	612.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.12	0.00	0.00	0.00	8,550,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	90.41	6.67	141.53	853.38	2,025,000.00 *gal*
TID Irr. Canal	Surface water	2.27	0.00	0.00	0.00	6,210,000.00 *gal*
Application event totals		92.67	6.67	141.53	853.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.12	0.00	0.00	0.00	8,550,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

Field #2 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	90.41	6.67	141.53	853.38	2,025,000.00 *gal*
TID Irr. Canal	Surface water	3.12	0.00	0.00	0.00	8,550,000.00 *gal*
Application event totals		93.53	6.67	141.53	853.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.12	0.00	0.00	0.00	8,550,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

Field #3 - 11/06/2022: Winter Forage

Field name: Field #3

Crop: Winter Forage

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	89.94	21.72	87.61	1,151.02	250.00 *ton*
Application event totals		89.94	21.72	87.61	1,151.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	58.31	7.44	84.99	499.30	270,000.00 *gal*
Application event totals		58.31	7.44	84.99	499.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	58.31	7.44	84.99	499.30	270,000.00 *gal*
Application event totals		58.31	7.44	84.99	499.30	

Field #3 - 05/14/2023: Corn, silage

Field name: Field #3

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.79	0.00	0.00	0.00	3,240,000.00 *gal*
Application event totals		3.79	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	110.61	10.81	181.90	1,144.00	630,000.00 *gal*
TID Irr. Canal	Surface water	3.47	0.00	0.00	0.00	2,970,000.00 *gal*
Application event totals		114.08	10.81	181.90	1,144.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.47	0.00	0.00	0.00	2,970,000.00 *gal*
Application event totals		3.47	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	90.01	6.65	140.90	849.59	630,000.00 *gal*
TID Irr. Canal	Surface water	3.05	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		93.06	6.65	140.90	849.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.05	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		3.05	0.00	0.00	0.00	

Field #3 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	90.01	6.65	140.90	849.59	630,000.00 *gal*
TID Irr. Canal	Surface water	3.47	0.00	0.00	0.00	2,970,000.00 *gal*
Application event totals		93.48	6.65	140.90	849.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.05	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		3.05	0.00	0.00	0.00	

Field #4 - 11/06/2022: Winter Forage

Field name: Field #4

Crop: Winter Forage

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	89.94	21.72	87.61	1,151.02	140.00 *ton*
Application event totals		89.94	21.72	87.61	1,151.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	60.74	7.75	88.53	520.10	157,500.00 *gal*
Application event totals		60.74	7.75	88.53	520.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/23/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	60.74	7.75	88.53	520.10	157,500.00 *gal*
Application event totals		60.74	7.75	88.53	520.10	

Field #4 - 05/14/2023: Corn, silage

Field name: Field #4

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	4.13	0.00	0.00	0.00	1,980,000.00 *gal*
Application event totals		4.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	98.76	9.65	162.41	1,021.43	315,000.00 *gal*
TID Irr. Canal	Surface water	3.76	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		102.52	9.65	162.41	1,021.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.76	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		3.76	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	91.84	6.78	143.77	866.93	360,000.00 *gal*
TID Irr. Canal	Surface water	3.19	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		95.03	6.78	143.77	866.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.19	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		3.19	0.00	0.00	0.00	

Field #4 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.40	4.24	89.86	541.83	225,000.00 *gal*
TID Irr. Canal	Surface water	3.19	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		60.59	4.24	89.86	541.83	

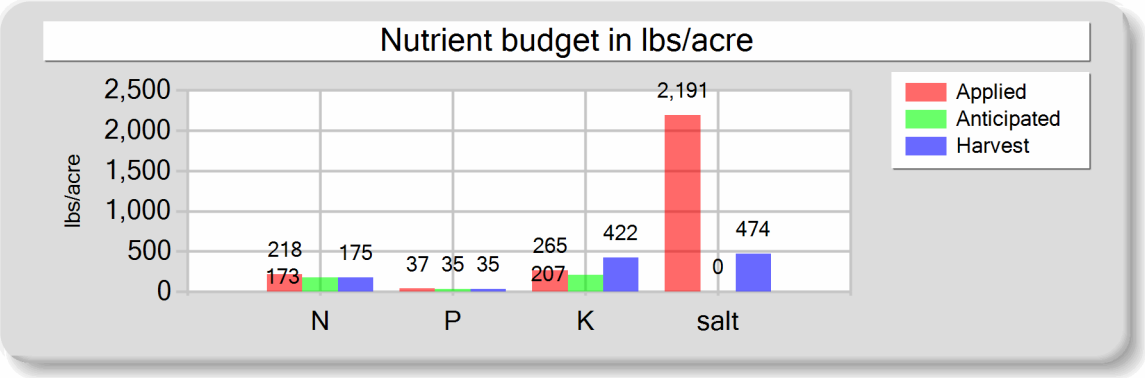
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	3.19	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		3.19	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #1 - 11/06/2022: Winter Forage

Field name: Field #1 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	89.94	21.72	87.61	1,151.02
Process wastewater	121.48	15.51	177.06	1,040.20
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.42	37.22	264.67	2,191.22
Anticipated crop nutrient removal	172.50	34.50	207.00	0.00
Actual crop nutrient removal	175.19	35.19	421.52	474.30
Nutrient balance	43.24	2.03	-156.85	1,716.92
Applied to removed ratio	1.25	1.06	0.63	4.62

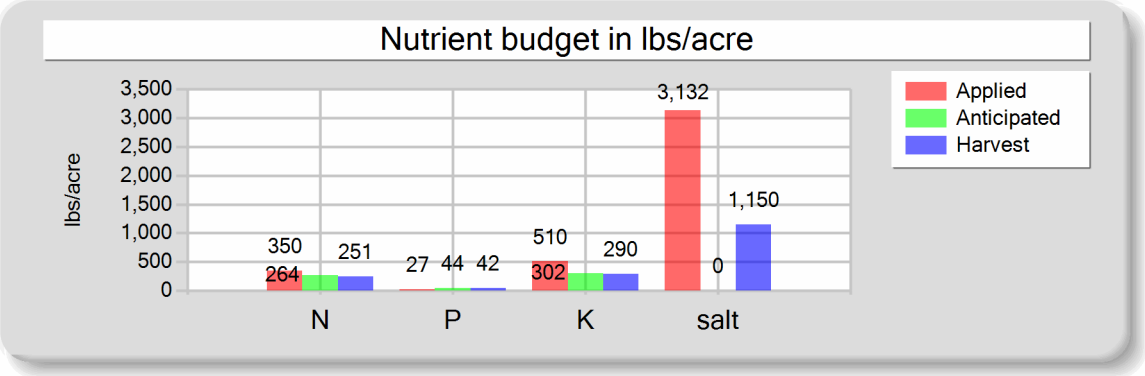
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
0.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #1 - 05/14/2023: Corn, silage

Field name: Field #1 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	319.20	26.71	510.09	3,131.88
Fresh water	24.10	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.29	26.71	510.09	3,131.88
Anticipated crop nutrient removal	263.90	43.50	301.60	0.00
Actual crop nutrient removal	250.54	42.04	290.00	1,149.72
Nutrient balance	99.76	-15.33	220.08	1,982.16
Applied to removed ratio	1.40	0.64	1.76	2.72

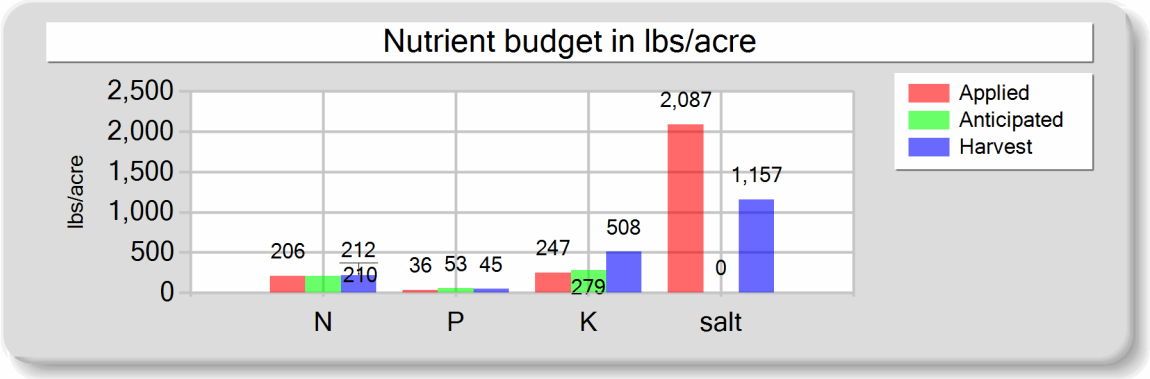
Fresh water applied	
9,900,000.00	*gallons*
364.58	*acre-inches*
30.38	*inches/acre*

Process wastewater applied	
990,000.00	*gallons*
36.46	*acre-inches*
3.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 11/06/2022: Winter Forage

Field name: Field #2 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	89.94	21.72	87.61	1,151.02
Process wastewater	109.33	13.96	159.35	936.18
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.27	35.67	246.96	2,087.20
Anticipated crop nutrient removal	210.00	52.50	279.00	0.00
Actual crop nutrient removal	212.11	44.61	507.75	1,157.41
Nutrient balance	-5.84	-8.93	-260.79	929.79
Applied to removed ratio	0.97	0.80	0.49	1.80

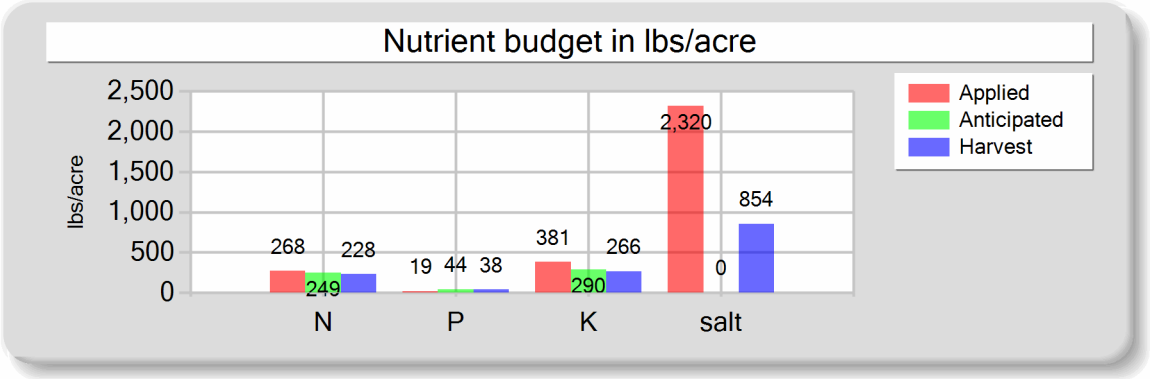
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
0.75	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #2 - 05/14/2023: Corn, silage

Field name: Field #2 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.07	19.14	380.50	2,319.62
Fresh water	21.06	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.14	19.14	380.50	2,319.62
Anticipated crop nutrient removal	249.40	43.50	290.00	0.00
Actual crop nutrient removal	228.28	37.63	265.71	854.10
Nutrient balance	39.85	-18.49	114.80	1,465.52
Applied to removed ratio	1.17	0.51	1.43	2.72

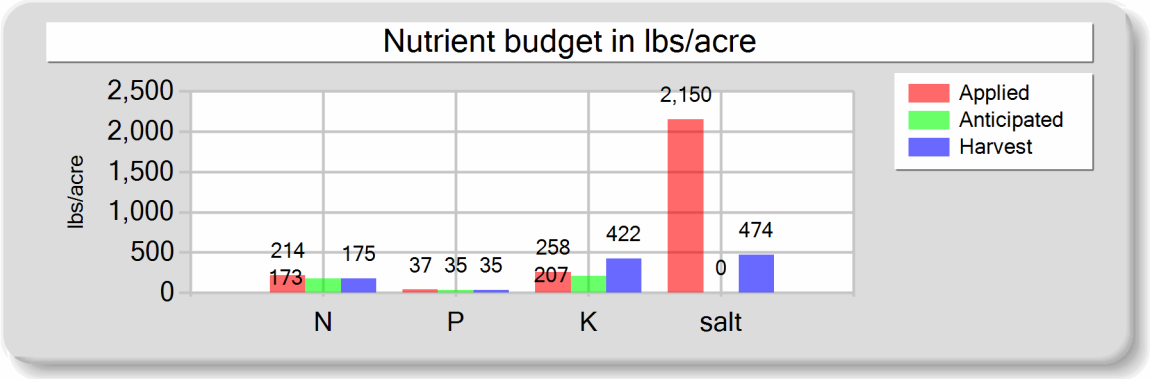
Fresh water applied	
57,690,000.00	*gallons*
2,124.53	*acre-inches*
26.56	*inches/acre*

Process wastewater applied	
5,130,000.00	*gallons*
188.92	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #3 - 11/06/2022: Winter Forage

Field name: Field #3 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	89.94	21.72	87.61	1,151.02
Process wastewater	116.62	14.89	169.98	998.60
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.56	36.60	257.59	2,149.62
Anticipated crop nutrient removal	172.50	34.50	207.00	0.00
Actual crop nutrient removal	175.19	35.19	421.52	474.30
Nutrient balance	38.38	1.41	-163.93	1,675.32
Applied to removed ratio	1.22	1.04	0.61	4.53

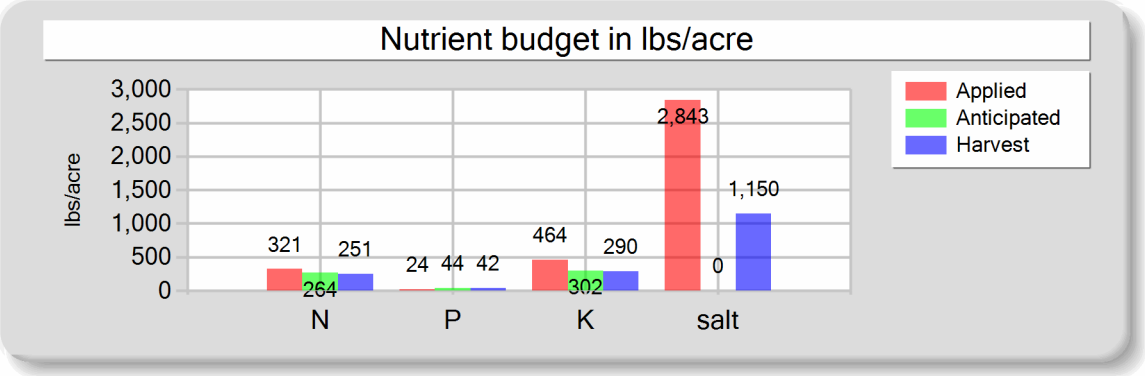
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
540,000.00 *gallons*
19.89 *acre-inches*
0.80 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #3 - 05/14/2023: Corn, silage

Field name: Field #3 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.63	24.10	463.70	2,843.17
Fresh water	23.34	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.97	24.10	463.70	2,843.17
Anticipated crop nutrient removal	263.90	43.50	301.60	0.00
Actual crop nutrient removal	250.54	42.04	290.00	1,149.72
Nutrient balance	70.43	-17.94	173.69	1,693.45
Applied to removed ratio	1.28	0.57	1.60	2.47

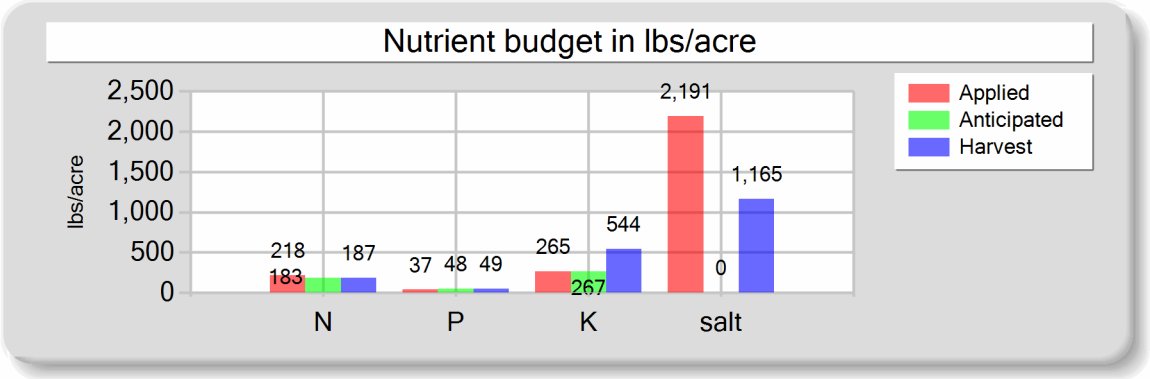
Fresh water applied	
19,980,000.00	*gallons*
735.80	*acre-inches*
29.43	*inches/acre*

Process wastewater applied	
1,890,000.00	*gallons*
69.60	*acre-inches*
2.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 11/06/2022: Winter Forage

Field name: Field #4 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	89.94	21.72	87.61	1,151.02
Process wastewater	121.48	15.51	177.06	1,040.20
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.42	37.22	264.67	2,191.22
Anticipated crop nutrient removal	183.00	48.00	267.00	0.00
Actual crop nutrient removal	187.40	48.70	544.20	1,164.64
Nutrient balance	31.02	-11.48	-279.53	1,026.59
Applied to removed ratio	1.17	0.76	0.49	1.88

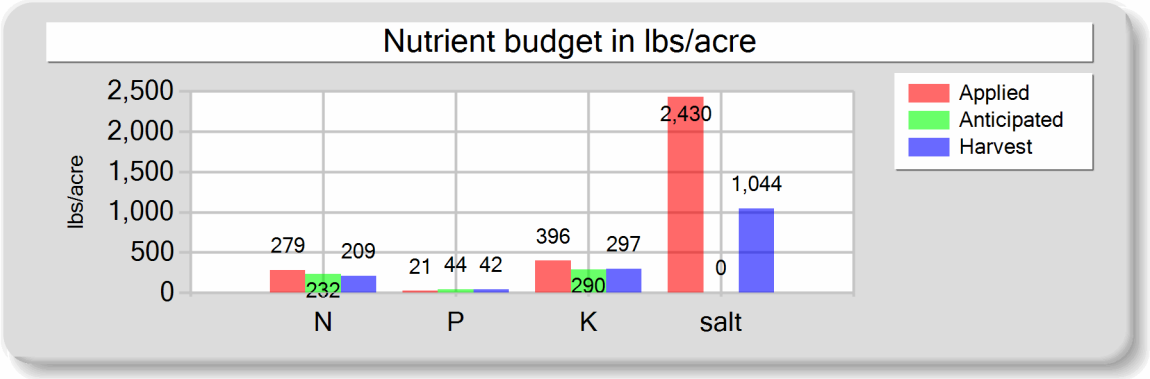
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
315,000.00 *gallons*
11.60 *acre-inches*
0.83 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #4 - 05/14/2023: Corn, silage

Field name: Field #4 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.01	20.67	396.04	2,430.18
Fresh water	24.41	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.42	20.67	396.04	2,430.18
Anticipated crop nutrient removal	232.00	43.50	290.00	0.00
Actual crop nutrient removal	208.78	41.92	297.11	1,043.90
Nutrient balance	70.64	-21.25	98.93	1,386.28
Applied to removed ratio	1.34	0.49	1.33	2.33

Fresh water applied	
11,700,000.00	*gallons*
430.87	*acre-inches*
30.78	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
2.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2350010D

Sample and source description: T2350010D

Sample date: 12/16/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	4,660.00	18,800.00	10,300.00	4,020.00	6,490.00	3,300.00	958.00		24.70
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132022B

Sample and source description: T3132022B

Sample date: 05/12/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	4,300.00	28,000.00							0.00
DL	1.00	1.00	1.00							1.00

T3258038D

Sample and source description: T3258038D

Sample date: 09/15/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,600.00	5,590.00	8,770.00	16,700.00	4,160.00	7,700.00	6,910.00	1,060.00		43.80
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3053004A

Sample and source description: T3053004A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	647.00	347.00	0.00	0.00	82.60	943.00								7,410.00	5,540
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132022A

Sample and source description: T3132022A

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	526.00	328.00	0.00	0.00	51.40	865.00								6,830.00	5,440
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3216013A

Sample and source description: T3216013A

Sample date: 08/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	428.00	255.00	0.00	0.00	31.60	670.00								6,160.00	4,040
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Ag Well

Ag Well

T3258038B

Sample description: T3258038B

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	44.50	0.00	44.50								1,370.00	0
DL	1.00	0.50	1.00								100.00	1

Domestic Well

T3258038C

Sample description: T3258038C

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.40	0.00	20.40								1,070.00	0
DL	1.00	0.50	1.00								100.00	1

TID Irr. Canal

T3258038A

Sample description: T3258038A

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.50	0.00	3.50								146.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

Field #1

T3258038I

Sample and source description: T3258038I

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			84.00		0.00	0.00	
DL			1.00		100.00	1.00	

Field #2

T3258035A

Sample and source description: T3258035A

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00		0.00	0.00	
DL			1.00		100.00	1.00	

T3258038J

Sample and source description: T3258038J

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			61.00		0.00	0.00	
DL			1.00		100.00	1.00	

Field #3

Field #3

T3258038I

Sample and source description: T3258038I

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			84.00		0.00	0.00	
DL			1.00		100.00	1.00	

Field #4

T3258035B

Sample and source description: T3258035B

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			41.00		0.00	0.00	
DL			1.00		100.00	1.00	

E. PLANT TISSUE ANALYSES

Field #1 - 11/06/2022: Winter Forage

T3132022C

Sample and source description: T3132022C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	4,600.00	55,100.00		6.20
DL	1.00	1.00	1.00		1.00

Field #1 - 05/14/2023: Corn, silage

Field #1 - 05/14/2023: Corn, silage

T3258038E

Sample and source description: T3258038E

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,450.00	16,900.00		6.70
DL	1.00	1.00	1.00		1.00

Field #2 - 11/06/2022: Winter Forage

T3132022D

Sample and source description: T3132022D

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	4,900.00	52,500.00		11.20
DL	1.00	1.00	1.00		1.00

T3132022E

Sample and source description: T3132022E

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	4,100.00	50,700.00		12.50
DL	1.00	1.00	1.00		1.00

Field #2 - 05/14/2023: Corn, silage

Field #2 - 05/14/2023: Corn, silage

T3258038F

Sample and source description: T3258038F

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,050.00	14,600.00		4.70
DL	1.00	1.00	1.00		1.00

T3258038G

Sample and source description: T3258038G

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,710.00	19,000.00		6.10
DL	1.00	1.00	1.00		1.00

Field #3 - 11/06/2022: Winter Forage

T3132022C

Sample and source description: T3132022C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	4,600.00	55,100.00		6.20
DL	1.00	1.00	1.00		1.00

Field #3 - 05/14/2023: Corn, silage

Field #3 - 05/14/2023: Corn, silage

T3258038E

Sample and source description: T3258038E

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,450.00	16,900.00		6.70
DL	1.00	1.00	1.00		1.00

Field #4 - 11/06/2022: Winter Forage

T3132022F

Sample and source description: T3132022F

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	4,600.00	51,400.00		11.00
DL	1.00	1.00	1.00		1.00

Field #4 - 05/14/2023: Corn, silage

T3258038H

Sample and source description: T3258038H

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,610.00	18,500.00		6.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

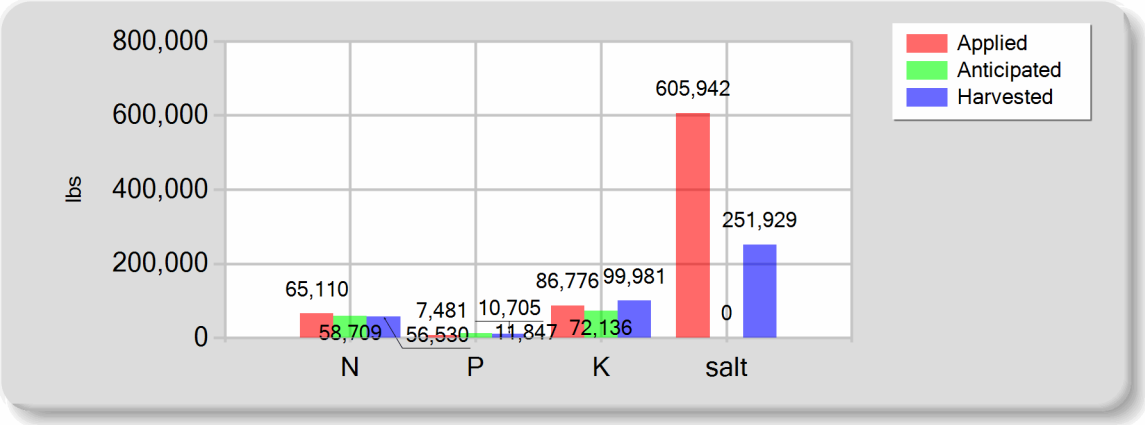
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

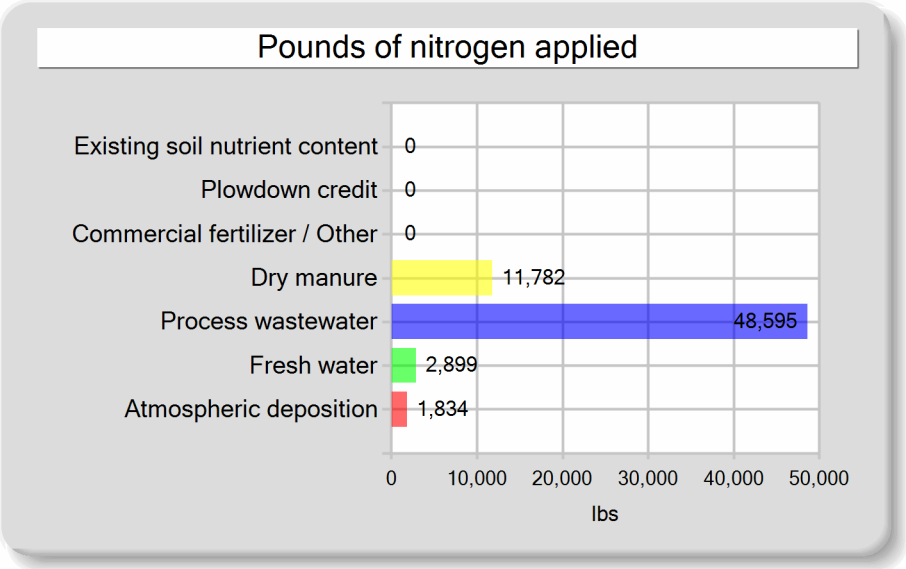
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

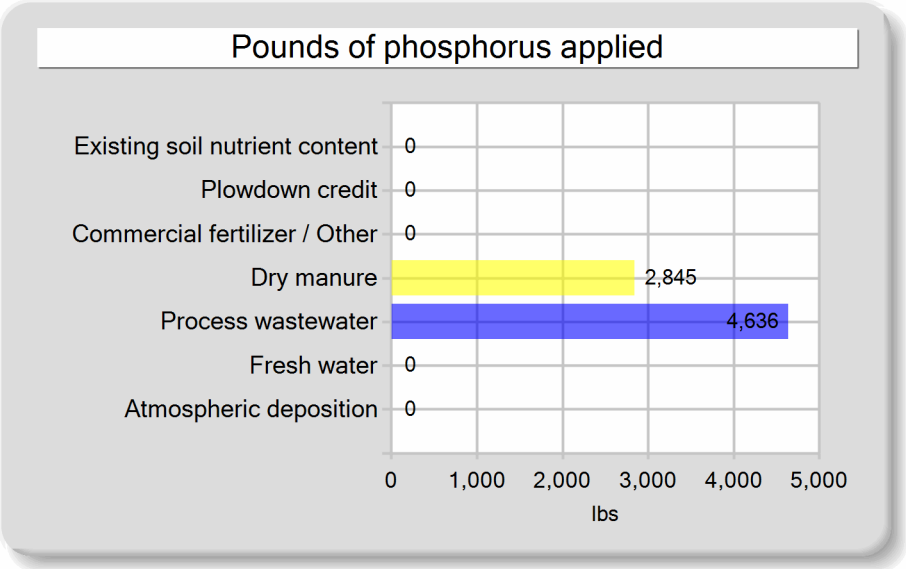
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	11,781.88	2,844.74	11,476.65	150,783.62
Process wastewater	48,594.81	4,635.78	75,299.52	455,158.83
Fresh water	2,899.43	0.00	0.00	0.00
Atmospheric deposition	1,834.00	0.00	0.00	0.00
Total nutrients applied	65,110.12	7,480.52	86,776.17	605,942.45
Anticipated crop nutrient removal	58,708.80	11,847.00	72,136.20	0.00
Actual crop nutrient removal	56,529.89	10,705.14	99,981.38	251,929.00
Nutrient balance	8,580.23	-3,224.62	-13,205.21	354,013.46
Applied to removed ratio	1.15	0.70	0.87	2.41

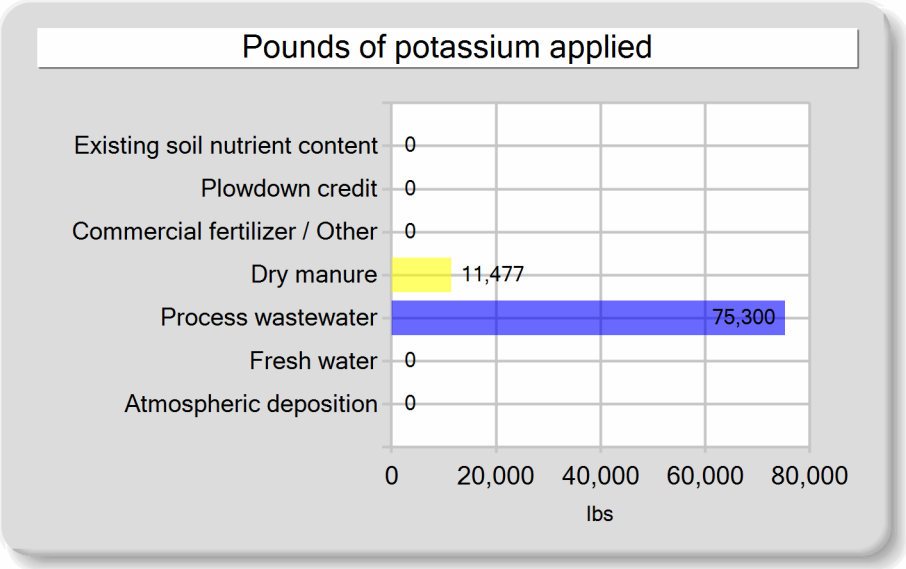
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

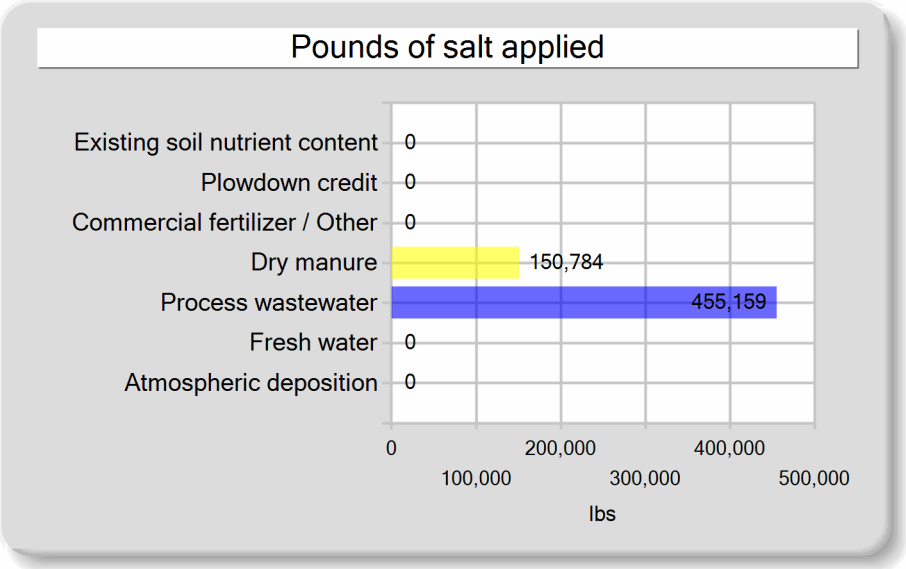


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Manuel Azevedo	Manuel Silva
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel & Maria Azevedo	Manuel & Maria Silva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-21-24	6-21-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Manuel Silva_____

Name of Dairy Facility: __Manuel & Maria Silva Dairy #2_____

Facility Address: _2800 White Rd._____Turlock_____95380__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Manuel Silva_____209-564-7297
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Holsapple's Manure Spreading_____

Address of Hauling Company /Person: _19085 Bradbury Rd._____Turlock_____95380_
Number and Street City Zip Code

Contact Person: __Rich Holsapple_____209-668-2400_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_John Alameda_____P.O. Box 862_____Hilmar____95324____209-495-7511__
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

___West of Lander Ave. on Fowler Rd.__Turlock__95380_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____5/23/23_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.59%
P 0.430% K 2.8%

Manure: ___1100_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___56.3%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: 100 Loads hauled

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 10/25/23

Hauler's Signature: _____ Date: 10/24/23

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Manuel Silva_____

Name of Dairy Facility: __Manuel & Maria Silva Dairy #2_____

Facility Address: _2800 White Rd._____Turlock_____95380__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Manuel Silva_____209-564-7297
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Holsapple's Manure Spreading_____

Address of Hauling Company /Person: _19085 Bradbury Rd._____Turlock_____95380_
Number and Street City Zip Code

Contact Person: __Rich Holsapple_____209-668-2400_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_John Alameda_____P.O. Box 862_____Hilmar____95324____209-495-7511__
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

___West of Lander Ave. on Fowler Rd.__Turlock__95380_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____10/17/23_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 2.66%
P 0.559% K 0.877%

Manure: ___1100_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___69.4%_____
Manure Density (if amount reported in cubic yards): _____

Waste Discharge Requirements General Order No. R5-2007-0035
Existing Milk Cow Dairies

Method used to determine amount of manure: 100 Loads hauled

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: [signature] Date: 10/25/23

Hauler's Signature: [signature] Date: 10/24/23

DENELE ANALYTICAL, INC.
865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

1/2

Client: Manuel & Maria Silva Dairy #2
Mailing address: 21487 White Rd.
(not facility address) Turlock, CA 95380
Phone#: (209)564-7297

Authorized Copy Released To: JCRK, INC.
(209)327-0992
Sampled By: Jacquie Captein
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/13/23	8:30a	TID Irr. Canal	Canal Irr.	21.6°		X				
2	✓	9:00a	Ag Well	Ag. Well	19.0°	20 min	X				
3	✓	9:30a	Domestic Well	Dom. Well	19.2°	✓	X				
4	✓	9:45a	Scraped Manure	Manure							M7
5	✓	10:00a	Fields 1+3	Corn Silage						X	
6	✓	10:10a	Field 2 North	✓						X	
7	✓	10:20a	Field 2 South	✓						X	
8	✓	10:30a	Field 4	✓						X	
9	✓	10:35a	Fields 1+3	Soil							K3
10	✓	10:45a	Field 2 North	✓							K3

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK		9/15/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	9-15-23 1:30			T3258038

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Manuel & Maria Silva Dairy #2 - Turlock
21487 White Rd.
Turlock, CA 95380

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Canal Water

NOTES: Temperature was above qualifying at 21.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
TID Irr. Canal	T3258038A	Electrical Conductivity (EC)	EPA 120.1 and	0.146	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.50	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	93.6	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
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21487 White Rd.
Turlock, CA 95380

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well	T3258038B	Electrical Conductivity (EC)	EPA 120.1 and	1.37	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	44.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	874	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

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Turlock, CA 95380

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well	T3258038C	Electrical Conductivity (EC)	EPA 120.1 and	1.07	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	20.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	684	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

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